Tabel MAPE Semua Model Prediksi Konsumsi Energi Hourly

57 konfigurasi | H1-H24 | hasil test asli | tidak ada training ulang

Tabel diurutkan berdasarkan Test Daily MAE. MAPE ditampilkan sebagai metrik pendukung dan perlu dibaca bersama MAE karena nilai aktual yang mendekati nol dapat memperbesar MAPE.

Rank	Model	Algoritma	Pendekatan	FTS	Daily MAPE (%)	Hourly MAPE (%)	Daily MAE (kWh)	Hourly MAE (kWh)
1	Residual FTS3-RandomForest	RandomForest	Residual	3	21.3101	47.7972	3.0977	0.2611
2	Hybrid FTS1-LightGBM Fixed 0.31	LightGBM	Hybrid	1	23.6967	35.4221	3.1354	0.2113
3	Residual FTS3-LightGBM	LightGBM	Residual	3	21.8002	47.4874	3.1380	0.2614
4	LightGBM Direct	LightGBM	Direct	-	23.6186	35.5359	3.1397	0.2120
5	Hybrid FTS1-LightGBM Validation threshold	LightGBM	Hybrid	1	23.6186	35.5359	3.1397	0.2120
6	Hybrid FTS2-LightGBM Validation threshold	LightGBM	Hybrid	2	23.6186	35.5359	3.1397	0.2120
7	Hybrid FTS3-LightGBM Validation threshold	LightGBM	Hybrid	3	23.6186	35.5359	3.1397	0.2120
8	Hybrid FTS3-LightGBM Fixed 0.31	LightGBM	Hybrid	3	23.6738	38.5994	3.1422	0.2216
9	Hybrid FTS3-RandomForest Validation threshold	RandomForest	Hybrid	3	23.3930	34.6604	3.1435	0.2064
10	RandomForest Direct	RandomForest	Direct	-	23.3930	34.6604	3.1435	0.2064
11	Hybrid FTS1-RandomForest Validation threshold	RandomForest	Hybrid	1	23.4409	34.6314	3.1482	0.2063
12	Hybrid FTS3-CatBoost Validation threshold	CatBoost	Hybrid	3	22.8633	35.2511	3.1493	0.2119
13	CatBoost Direct	CatBoost	Direct	-	22.8633	35.2511	3.1493	0.2119
14	Hybrid FTS3-RandomForest Fixed 0.31	RandomForest	Hybrid	3	23.5378	35.7394	3.1591	0.2097
15	Hybrid FTS1-RandomForest Fixed 0.31	RandomForest	Hybrid	1	23.6604	34.7759	3.1708	0.2067
16	Hybrid FTS2-CatBoost Validation threshold	CatBoost	Hybrid	2	22.5570	53.3961	3.1738	0.2719
17	Hybrid FTS1-CatBoost Validation threshold	CatBoost	Hybrid	1	23.3021	35.6906	3.1808	0.2131
18	Hybrid FTS3-CatBoost Fixed 0.31	CatBoost	Hybrid	3	23.1918	37.3977	3.1826	0.2186
19	SARIMA (1, 0, 1) (1, 0, 1, 24)	-	SARIMA	-	20.6933	31.9491	3.1841	0.2053
20	Hybrid FTS1-CatBoost Fixed 0.31	CatBoost	Hybrid	1	23.2194	35.3279	3.1855	0.2120

Rank	Model	Algoritma	Pendekatan	FTS	Daily MAPE (%)	Hourly MAPE (%)	Daily MAE (kWh)	Hourly MAE (kWh)
21	Hybrid FTS2-CatBoost Fixed 0.31	CatBoost	Hybrid	2	23.3464	39.8967	3.2014	0.2267
22	Residual FTS1-LightGBM	LightGBM	Residual	1	23.5029	36.7858	3.2312	0.2152
23	Hybrid FTS2-LightGBM Fixed 0.31	LightGBM	Hybrid	2	24.4030	42.2542	3.2402	0.2337
24	ARIMA (2, 1, 1) None	-	ARIMA	-	23.1124	48.5165	3.2781	0.2696
25	Hybrid FTS2-RandomForest Validation threshold	RandomForest	Hybrid	2	24.1211	52.5242	3.2787	0.2653
26	Residual FTS1-RandomForest	RandomForest	Residual	1	23.9337	37.6338	3.2952	0.2198
27	ARIMA (1, 1, 1) None	-	ARIMA	-	23.9042	50.7815	3.3111	0.2754
28	Hybrid FTS3-SVR Validation threshold	SVR	Hybrid	3	21.0447	29.6960	3.3514	0.2123
29	Residual FTS1-CatBoost	CatBoost	Residual	1	25.1784	39.9907	3.3564	0.2288
30	Hybrid FTS2-RandomForest Fixed 0.31	RandomForest	Hybrid	2	25.7887	39.5646	3.3624	0.2218
31	Hybrid FTS3-XGBoost Fixed 0.31	XGBoost	Hybrid	3	24.7099	39.1052	3.3708	0.2249
32	SVR Direct	SVR	Direct	-	21.0947	28.7320	3.3821	0.2088
33	Hybrid FTS3-XGBoost Validation threshold	XGBoost	Hybrid	3	24.8636	37.3247	3.3885	0.2193
34	XGBoost Direct	XGBoost	Direct	-	24.8636	37.3247	3.3885	0.2193
35	Residual FTS3-CatBoost	CatBoost	Residual	3	23.5234	48.5158	3.3912	0.2646
36	Hybrid FTS1-XGBoost Fixed 0.31	XGBoost	Hybrid	1	25.0363	37.1843	3.4014	0.2188
37	Hybrid FTS1-XGBoost Validation threshold	XGBoost	Hybrid	1	25.0272	37.1852	3.4052	0.2188
38	Residual FTS1-XGBoost	XGBoost	Residual	1	25.3999	39.8630	3.4082	0.2294
39	Hybrid FTS2-XGBoost Validation threshold	XGBoost	Hybrid	2	25.1511	48.2279	3.4294	0.2548
40	Residual FTS3-XGBoost	XGBoost	Residual	3	23.8222	49.8134	3.4301	0.2721
41	Hybrid FTS3-SVR Fixed 0.31	SVR	Hybrid	3	21.9519	33.6971	3.4330	0.2251
42	Hybrid FTS2-XGBoost Fixed 0.31	XGBoost	Hybrid	2	25.4227	41.2220	3.4572	0.2317
43	Weighted FTS order 3	-	-	-	25.3629	53.7926	3.6097	0.2933
44	Hybrid FTS1-SVR Validation threshold	SVR	Hybrid	1	23.2141	31.7097	3.6687	0.2225

Rank	Model	Algoritma	Pendekatan	FTS	Daily MAPE (%)	Hourly MAPE (%)	Daily MAE (kWh)	Hourly MAE (kWh)
45	Hybrid FTS1-SVR Fixed 0.31	SVR	Hybrid	1	23.2141	31.7097	3.6687	0.2225
46	Residual FTS3-SVR	SVR	Residual	3	26.2272	47.4140	3.8651	0.2739
47	Hybrid FTS2-SVR Fixed 0.31	SVR	Hybrid	2	26.0091	36.1110	3.8979	0.2325
48	Weighted FTS order 2	-	-	-	35.7298	60.8947	4.3668	0.3170
49	Hybrid FTS2-SVR Validation threshold	SVR	Hybrid	2	31.6487	44.1546	4.4689	0.2590
50	Residual FTS1-SVR	SVR	Residual	1	29.3990	37.1465	4.5065	0.2488
51	Residual FTS2-CatBoost	CatBoost	Residual	2	40.4654	54.6969	5.2640	0.2971
52	Residual FTS2-XGBoost	XGBoost	Residual	2	43.4967	57.4090	5.5791	0.3075
53	Residual FTS2-RandomForest	RandomForest	Residual	2	45.3961	57.6632	5.7943	0.3078
54	Weighted FTS order 1	-	-	-	49.4256	83.1581	5.8303	0.3697
55	Residual FTS2-LightGBM	LightGBM	Residual	2	47.2437	61.3853	5.9018	0.3205
56	Residual FTS2-SVR	SVR	Residual	2	51.5011	61.5494	6.5084	0.3347
57	SARIMA (1, 1, 1) (1, 0, 1, 24)	-	SARIMA	-	93.4122	93.6096	12.8550	0.5356

MAPE harian terendah: SARIMA (1, 0, 1) (1, 0, 1, 24) (20.6933%). MAPE hourly terendah: SVR Direct (28.7320%). Pemilihan model penelitian tetap harus mengacu pada validation, bukan hanya MAPE test.